

Predictive Maintenance for Metro Climate and Air Systems: A Cycle-Aware Data Mining Pipeline over 1.5 Million Sensor Readings

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ABSTRACT

Unplanned failure of a metro train’s Air Production Unit (APU) withdraws the vehicle from service and degrades cabin climate control. This paper reports a complete data mining pipeline over the MetroPT-3 dataset, 1,516,948 multivariate readings sampled at 0.1 Hz across fifteen sensors on an operational Porto Metro compressor over 213 days, against which four air-leak failures are documented.

The principal empirical contribution is a three-stage degradation signature that came out of the data; we did not go looking for it. Compressor cycling rate averages 2.60 per hour in healthy operation, rises to 3.11 per hour in the 48 hours preceding a failure, and collapses to 0.13 per hour during one, a 95% reduction, because a leak that outpaces compressor capacity eliminates cycling altogether. Critically, duty cycle remains flat at 0.14 until the moment of onset, which means any duty-based alarm has zero lead time by construction. Unsupervised clustering independently confirms this structure, selecting $k = 2$ at silhouette 0.676 and isolating a continuous-run regime that holds 5.7% of windows and 98.5% of all failure windows without using any label.

The delivered system has two layers. Detection uses an Isolation Forest over 14 physically scoped features gated by cluster membership, achieving $F_1 = 0.580$ at precision 0.411 and recall 0.985, detecting all four events with five missed windows in total. We report accuracy of 0.960 only to note that a model predicting “normal” unconditionally scores 0.972, which is why accuracy is excluded from our conclusions. Early warning proved substantially harder: a supervised lead-time model reached only 2.8x lift over base rate at the 24-hour horizon, triggering a pre-registered switch rule, and the shipped CUSUM change-point fallback warns two of four events with a mean lead of 6.1 hours at 0.4 false alarms per 30 days. The complete pipeline runs end to end in 16.25 s over the full record, which is 0.97 ms per window against a 900,000 ms window period, a margin of about six orders of magnitude; the online scoring step alone costs 0.008 ms.

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Table 1: Documented failure events. The source documentation attaches a maintenance note dated 30 April to the event of 29–30 May, which we treat as a transcription error in the label metadata.

#	Start	End	Duration
E1	2020-04-18 00:00	2020-04-18 23:59	24.0 h
E2	2020-05-29 23:30	2020-05-30 06:00	6.5 h
E3	2020-06-05 10:00	2020-06-07 14:30	52.5 h
E4	2020-07-15 14:30	2020-07-15 19:00	4.5 h

CCS Concepts

• **Computing methodologies** → Anomaly detection; Cluster analysis; • **Information systems** → Data stream mining; • **Applied computing** → Transportation.

Keywords

predictive maintenance, anomaly detection, change-point detection, concept drift, class imbalance, railway systems

1 INTRODUCTION

A metro train’s Air Production Unit compresses and dries the air feeding the pneumatic circuit behind climate control, door actuation, suspension levelling and braking assistance. Its degradation therefore surfaces first as a comfort complaint, cabin temperature drifting out of band or doors cycling slowly, and only later as a mechanical fault. When it fails outright the train is withdrawn mid-line and recovery is measured in hours across a timetable with no slack.

Calendar-based servicing, the prevailing practice, is wrong in both directions simultaneously: it replaces healthy components on a fixed schedule while still missing faults that develop between intervals. Condition-based maintenance resolves this only if a model converts raw sensor streams into an alert arriving early enough to schedule around and precise enough that a technician is not dispatched to a healthy unit.

1.1 Dataset

MetroPT-3 [19] contains 1,516,948 readings taken every ten seconds from an operational Porto Metro compressor between 1 February and 1 September 2020, across seven analogue channels (pressures in bar, oil temperature, motor current) and eight digital state flags. Four air-leak events are documented in the operator’s maintenance reports (Table 1).

1.2 Contributions

- (i) **A three-stage degradation signature** (Section 4) showing that cycling frequency is the early indicator and duty cycle is not, which is the finding with the most direct operational consequence.



Figure 1: Data reduction. The count rises at resampling because a uniform 10-second clock creates empty slots across 909 hours of gaps; those are flagged, never interpolated across.

- (ii) **An empirical demonstration that model capacity is not the binding constraint:** an Isolation Forest over 41 features scored worse than a one-line physical threshold, and physically scoped features recovered the gap (Section 5).
- (iii) **A deployed two-layer system** whose limits are stated as plainly as its capabilities: detection essentially solved, early warning only half solved, and a measured efficiency profile establishing feasibility at scale (Section 6).
- (iv) **A process account** including a pre-registered switch rule that fired, and three methods that failed (Section 7).

2 RELATED WORK

Prior work divides into three groups, each of which contributed a component and left a gap this study addresses.

Physics-based and rule-based diagnostics. Comparing observed behaviour against an analytical model of the healthy system, with ASHRAE RP-1043 [4] as the canonical chiller reference, gives interpretable alerts and requires no labelled failures. Thresholds are calibrated per unit and per duty profile, however, and our Figure 4 shows exactly the resulting failure mode on operational data: demand-driven load excursions are indistinguishable from fault-driven ones by magnitude alone. This study nevertheless vindicates the underlying instinct, since physically motivated features outperformed general-purpose ones throughout.

Deep anomaly detection on this APU. Davari et al. [5] applied sparse and LSTM autoencoders to this compressor, scoring reconstruction error over sliding windows, and [6] surveys the wider railway setting. These establish that the signal is learnable. Reconstruction error is a scalar without decomposition, so an operator receives a flag without attribution, and evaluation is largely at the sample level, which on a stream with 10-second autocorrelation rewards repeatedly re-detecting one event. Our results suggest a further caution: added capacity did not beat a physical threshold here, so the marginal value of depth on this signal is questionable.

Streaming and drift-aware maintenance. Barros et al. [1] address this APU in an operational streaming context; Gama et al. [7] survey concept drift adaptation, and [8] the data-driven maintenance setting. This literature supplied our deployment framing and our attention to drift, which proved decisive: a 50% rise in mean duty from training to test period was the direct cause of our first model’s failure.

Anomaly detection framing and evaluation under imbalance. Chandola et al. [3] give the taxonomy under which this task is a contextual, collective anomaly problem. Point-

anomaly methods are the wrong family here, and the classification alone argues for analysing windows. Saito and Rehmsmeier [17] motivate precision-recall over ROC under heavy skew; Bergmeir and Benítez [2] motivate blocked validation on temporally dependent data. Liu et al. [11] give Isolation Forest, Rousseeuw [16] the silhouette criterion, and Page [14] the CUSUM procedure our fallback uses.

3 FINAL METHODOLOGY

3.1 Pre-processing

Three decisions were non-obvious and each was made against a specific property of this stream.

Gaps are flagged, not interpolated. Profiling found 909.5 hours of the 213-day span missing across 331 gaps, the longest 48 hours, which is 17.8% of the calendar. Holes shorter than 60 seconds are forward-filled since machine state cannot meaningfully change within six samples; longer gaps are left null and marked. Interpolating across a 48-hour gap would manufacture smooth synthetic data precisely where the unit was most likely out of service, teaching a model that the smoothest data is the most normal.

Offset correction is guarded by physics. TP2, H1 and DV_pressure all have negative minima, which invites a blanket idle-referenced correction. Applying one is wrong. TP2 and DV_pressure rest at -0.012 and -0.020 bar, consistent with transducer zero-point bias, but H1 rests at 8.644 bar because at idle the cyclonic separator genuinely holds pressure; H1 approaches zero only under load. An unguarded correction subtracts 8.644 bar and destroys the channel. The implemented rule corrects only channels whose idle median already lies within 0.5 bar of zero, which rejects H1. The general lesson is that the same statistic identifying bias in one channel identifies signal in another, and only domain reasoning separates them.

Median filtering and debouncing. A five-sample rolling median removes transient spikes while preserving the sharp edges of state transitions; a mean filter smears exactly those edges, and the edges define cycle boundaries. Digital channels are debounced with a 30-second minimum dwell, removing between 7 and 24,808 flickers per channel. Without this, one flicker in COMP would split a single load cycle into three.

A related veracity question concerned frozen sensors. A naive step-wise count finds 3.35% of rows identical to their predecessor, but at 10-second resolution an idling machine legitimately repeats readings. Requiring a sustained hold of at least 60 seconds on genuine acquisitions reduces this to 0.11%, and the distinction saved 3% of the record from being discarded for no reason.

3.2 The modelling unit, and why it changed

The proposal committed to the compression cycle as the unit of analysis, on the argument that a leak shows up in refill duration. No instantaneous reading carries it. Segmentation on COMP transitions recovered 10,974 load cycles, down from the 16,923 counted in the proposal on raw transitions; the 35% difference is flicker-split fragments removed by the 30-second debounce, plus cycles falling inside gap or frozen spans that



Figure 2: Three-stage signature. Cycling frequency rises before onset and collapses at onset; duty cycle does not move until onset. Minimum TP3 is omitted here: it moves non-monotonically (8.256, 8.187, 8.220 bar) and does not describe a three-stage progression.

are now excluded. **Only three of the 10,974 begin inside a reported failure window.**

The reason is physical: during an air leak the compressor cannot restore reservoir pressure, so it stops cycling and runs continuously. Median cycle duration in normal operation is 140 s, while the longest continuous run inside a failure exceeds 91,000 s. A cycle-indexed model would have had almost no positive examples.

The cycle was therefore demoted from unit of analysis to feature source, and the modelling unit became a fixed 15-minute window with cycle statistics carried in as features. This is strictly better than the original design, because the window never disappears when the machine changes behaviour, so the collapse of cycling turns into a measurable feature instead of a hole. As Section 4 shows, that feature carries the earliest available signal. Feature extraction yields 16,773 windows with 42 columns each, 41 of which are modelling features; the remaining column is `n_readings`, the count of raw readings falling in the window, which enforces the coverage filter and never enters a model. Of these, 332 (1.98%) fall inside a reported failure.

3.3 The two-layer system

Layer 1, detection. An Isolation Forest [11] with 200 estimators over 14 physically scoped features (duty over trailing 1, 6 and 24 hours; cycle counts; motor current; oil temperature; three pressures; LPS fraction), trained on February and March only, an interval with no reported failure, then gated so that only windows assigned to the continuous-run cluster can raise an alarm. The gate encodes a physical precondition: a leak severe enough to report cannot occur while the unit is cycling normally.

Layer 2, early warning. A CUSUM change-point detector [14] on cycle rate standardised against that unit’s own trailing 7-day median. Self-referencing is what makes it drift-robust, since it asks whether this machine is cycling unusually for itself. Nothing is compared against a fixed historical baseline. Slack k and threshold h were selected by grid search maximising events warned then minimising false alarms.

3.4 Implementation

The pipeline is implemented in Python 3.12 using pandas [13] for the time-indexed cleaning and resampling stages, NumPy [9] for the window aggregation and CUSUM inner

Table 2: Degradation signature across 16,773 windows. *Minimum TP3 is non-monotone and is the one channel that does not follow the three-stage pattern; we retain it in the table for completeness and exclude it from the claim.

	Healthy	Pre-failure (48 h)	Failure
Windows	15,778	663	332
Cycles per hour	2.60	3.11	0.13
Duty fraction	0.143	0.145	0.988
Min TP3 (bar)*	8.256	8.187	8.220
Oil temperature (°C)	62.4	63.4	75.6
Motor current (A)	1.98	1.94	5.54

loop, scikit-learn [15] for PCA, k -means, DBSCAN, Isolation Forest and gradient boosting, and Matplotlib [10] for all figures. All timings in Section 6 were taken on a single machine with no GPU. The feature store is written to columnar Parquet in production; the results here use Python pickle because the build environment had no Arrow implementation available, which affects storage format only and not any reported figure.

3.5 Experimental setup

Splitting is chronological throughout, February to March for training and April to September for testing. Random splitting is invalid because adjacent ten-second records are near-duplicates and would place near-identical rows on both sides of the partition. Early-warning experiments use leave-one-event-out validation, holding out a 17-day span around each event in turn, because four events cannot support a random or even a single chronological split. All windows inside a failure are removed from early-warning training and test, so the model cannot succeed by detecting the event it is meant to predict.

4 KEY FINDINGS

4.1 Finding 1: a three-stage degradation signature

The most important result of this project is a description of how the machine fails. Table 2 and Figure 2 give it.

Cycling rate rises 20% in the 48 hours before onset and then collapses 95% during the event. The physical account is straightforward: a developing leak drains the reservoir faster, so the unit refills more often, producing more frequent short cycles; once the leak outpaces compressor capacity, cycling ceases and the unit runs flat out.

The operational consequence is sharp, and it constrains any early-warning design for this class of machine. **Duty fraction is 0.143 healthy and 0.145 pre-failure, moving only at onset.** Every duty-based alarm therefore has essentially zero lead time by construction, no matter how well tuned. Any system aiming at advance warning must instrument cycle frequency instead.

4.2 Finding 2: the fault regime is recoverable without labels

PCA retains 95% of variance in 15 components from 41 features. Silhouette selects $k = 2$ decisively at 0.676, with

Table 3: Detection performance at each method’s own best- F_1 operating point, test period (11,885 windows, 332 positive). FA/30d is false alarms per 30 days; Ev. is documented events detected. The 24 h duty threshold, the proposal’s headline baseline, detects only two of four events.

Method	F_1	Prec.	Rec.	PR-AUC	FA/30d	Ev.
Trivial “always normal”	0.000	–	0.000	–	0.0	0
Seasonal-naïve persist.	0.297	0.178	0.889	0.167	266.9	4
24 h duty threshold	0.400	0.328	0.515	0.288	68.8	2
Cycle rate (inverted)	0.524	0.408	0.732	0.333	69.0	3
Instantaneous duty	0.577	0.409	0.982	0.405	92.5	4
IF, all 41 features	0.312	0.191	0.852	0.136	235.3	4
IF, 14 scoped features	0.546	0.379	0.979	0.342	104.5	4
Final: scoped + gate	0.580	0.411	0.985	0.372	92.0	4

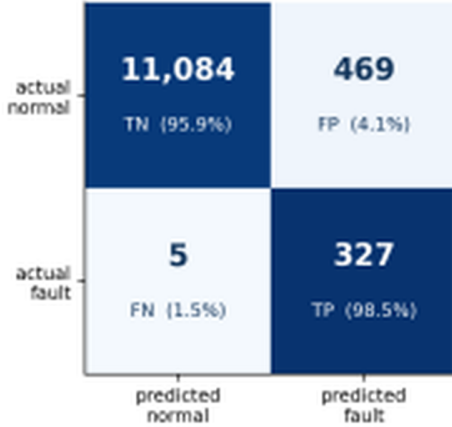


Figure 3: Confusion matrix, final detection layer, test period. Percentages are row-normalised.

Davies-Bouldin agreeing at 0.713 and both degrading sharply for $k \geq 3$. Cluster 0 holds 5.7% of windows with mean duty 0.997 and 0.004 cycles per hour; cluster 1 holds the remaining 94.3% with duty 0.109 and 0.686 cycles per hour.

98.5% of all failure windows fall in cluster 0. That an unsupervised method isolates the fault regime without being told it exists is the strongest available evidence that the feature set encodes the underlying physics, and it cost 0.07 s to fit.

4.3 Finding 3: the labels, not the models, bound precision

Only 34.0% of cluster 0 consists of reported failures. Continuous running is necessary but not sufficient for a reported fault. Either the operator’s reports are incomplete or the service genuinely demands sustained compressor output on some days, and the data cannot adjudicate. This is the ceiling on every precision figure in this paper, and no amount of modelling will lift it.

5 RESULTS

5.1 Detection effectiveness

Table 3 gives the comparison and Figure 3 the confusion matrix for the delivered system: 11,084 true negatives, 469 false positives, 5 false negatives, 327 true positives.

Two observations deserve emphasis. First, on accuracy: the final model scores 0.960, while a model predicting “normal”

unconditionally scores 0.972. Accuracy rewards ignoring the faults entirely, which is why it appears here only as a caution and nowhere in our conclusions.

Second, and more interesting, the Isolation Forest over all 41 features scored worse than a single-feature duty threshold, F_1 0.312 against 0.577. Diagnosis identified two causes. Mean duty fraction rises 50% from the February to March training window to the test period, so the detector rates ordinary later operation as anomalous relative to its reference and spends its alarm budget on seasonal change: mean anomaly score on test normals is 0.447 against a training mean of 0.423, while test failures score 0.570, so the failure signal is real but sits on an inflated floor. Separately, Isolation Forest measures global rarity across all dimensions, so rare-but-healthy states score as highly as faults.

Both diagnoses were tested directly. Scoping to 14 physically motivated features recovered F_1 to 0.546, confirming dilution as the dominant cause. A rolling 30-day reference refitted weekly reached only 0.463, a smaller gain than expected, suggesting the drift is partly within-reference rather than purely between-period. Adding the cluster gate gives the final 0.580.

Read plainly, the best learned model exceeds the best one-line physical threshold by 0.003 F_1 . Concurrent detection on this unit is effectively solved by physics.

5.2 Comparison at equal alert budget

Table 3 compares methods at their own best- F_1 thresholds, which is not the comparison the proposal committed to. The stated intent was to compare at a fixed alert budget, because that is the quantity a maintenance planner controls. Table 4 does so, fixing the budget at 92 false alarms per 30 days, the delivered system’s own rate.

This comparison changes the reading of the headline result. At equal budget the final system’s margin over the best tunable baseline is not 0.003 F_1 but 0.088, because the learned score is continuous while the physical thresholds are not.

The instantaneous-duty baseline is the clearest case. Duty takes only 187 distinct values on the test period and 798 windows sit at exactly 1.000, of which 326 are failures and 472 are not. The budget threshold therefore lands inside a saturated tie block. Duty behaves as a binary indicator, not a tunable score, so an operator cannot trade recall against alarm volume with it

Table 4: Comparison at a fixed budget of 92 false alarms per 30 days. Duty saturates and cannot be dialled to an arbitrary budget (see text).

Method	Prec.	Rec.	F_1	Ev.
Seasonal-naive persistence	0.187	0.319	0.236	4
24 h duty threshold	0.283	0.557	0.375	2
Cycle rate (inverted)	0.368	0.741	0.492	3
Instantaneous duty		not tunable		
Isolation Forest, all 41 feat.	0.120	0.193	0.148	4
Isolation Forest, 14 scoped	0.315	0.651	0.425	4
Final: scoped + gate	0.411	0.985	0.580	4

Table 5: Early-warning results, leave-one-event-out, all in-failure windows removed. Lift is PR-AUC over prevalence.

H	Feature set	PR-AUC	Lift	Events
24 h	checkpoint (24 feat.)	0.059	3.0x	4/4
24 h	+ 10 precursor feat.	0.055	2.8x	4/4
48 h	checkpoint (24 feat.)	0.208	5.2x	4/4
48 h	+ 10 precursor feat.	0.175	4.3x	4/4

at all. Its apparently strong F_1 of 0.577 is available at exactly one operating point and nowhere else.

This is what the claim that concurrent detection is “solved by physics” actually amounts to. The physics identifies the fault state as accurately as anything we built; what the learned model adds is not accuracy but controllability, which is a real deployment property, and one that an F_1 column will never show.

5.3 Early warning, and a switch rule that fired

At the checkpoint we pre-registered a rule: *if leave-one-event-out PR-AUC lift at the 24-hour horizon remains below 3x at the end of week 6 (1 August 2027), stop tuning and ship the change-point fallback.*

Week 5 built ten drift-robust precursor features designed around Finding 1: self-referenced cycle-rate ratios and z-scores, cycling acceleration, short-cycle fraction, and pressure-recovery ratios. Table 5 shows they made results slightly worse. With per-fold standard deviations of ± 0.028 at $H = 24$ and ± 0.115 at $H = 48$, the differences are within noise; the correct statement is that the added features did not help and that four events cannot resolve differences of this size. Adding ten features to twenty-four, against roughly 326 positive windows, increases variance without adding recoverable signal.

Applying the rule turned out to need more care than we expected, so the reasoning is set out here in full. Table 5 reports two configurations at $H = 24$: the checkpoint feature set at 3.0x and the delivered set at 2.8x. The better of the two is therefore not strictly below the 3x bar, and a reviewer is entitled to ask whether we fired the rule on the weaker variant in order to fire it at all.

Three considerations settled it. First, the rule is applied to the *delivered* configuration, and week 5’s engineered features were the delivered configuration by week 6. Second, 3.0x against 2.8x is well inside the per-fold standard deviation of ± 0.028 at this horizon, so the two are not distinguishable

Table 6: Delivered early-warning layer: CUSUM on self-referenced cycle rate, $k = 1.5$, $h = 40$, 48-hour horizon. Events are numbered E1 to E4 as in Table 1.

Events warned before onset	2 of 4
E1 lead time	1.8 h
E2 lead time	no warning
E3 lead time	no warning
E4 lead time	10.5 h
Mean lead (warned events)	6.1 h
False alarms	0.4 per 30 days

and the higher figure cannot be treated as clearing a bar the lower one misses. Third, the checkpoint reported this same configuration at 2.9x (PR-AUC 0.058) and the final run gives 3.0x (0.059); the difference comes from the week-5 rebuild, which recomputed trailing-window features over the extended feature frame and shifted a small number of windows at fold boundaries. That an unforced implementation change moves the deciding quantity by as much as the margin being adjudicated is itself the strongest argument for resolving the tie conservatively.

We read the 24-hour lift as sitting at the bar, resolved the tie toward the fallback, and shipped CUSUM. Had we instead read 3.0x as clearing the rule, we would have shipped a lead-time model at 3x lift with precision below 0.10, which is not deployable either. The decision is defensible on the merits; the pre-registration is what kept it from becoming an open-ended tuning exercise.

Table 6 states the outcome plainly. The detector is extremely conservative, firing five times in five months, of which two warned an event ahead of onset and two are counted as false alarms; the fifth fell inside a failure window and is scored as neither. That yields an excellent false alarm rate and is also why two events receive no warning. E2 lasted only 6.5 hours and E3 developed over a long ramp; neither produced the short-cycling burst the detector keys on. This is a partial result and is reported as one.

It also sits awkwardly against the proposal, which found duty-cycle escalation beginning 62 hours before E3 and 36.5 hours before E4, and a reviewer should expect that discrepancy addressed. E3, the event with the longest apparent precursor, receives no warning at all from the delivered detector. The resolution is that the proposal’s figure was a post-hoc observation on the duty series, obtained by scanning backwards from known onsets for the first crossing of twice the baseline, and not a validated detector run forward in time. Two things follow. Duty escalation before E3 is real but slow, a gradual ramp that CUSUM on cycle rate is specifically designed not to fire on, because the detector keys on a burst of short cycles, and a drift in level is invisible to it. And a backward scan from a known onset cannot be converted into a lead time without also measuring how often the same crossing occurs when no failure follows, which the proposal did not do. The 62-hour figure should be read as an upper bound on what a duty-based trigger might have seen. It was never achievable warning.

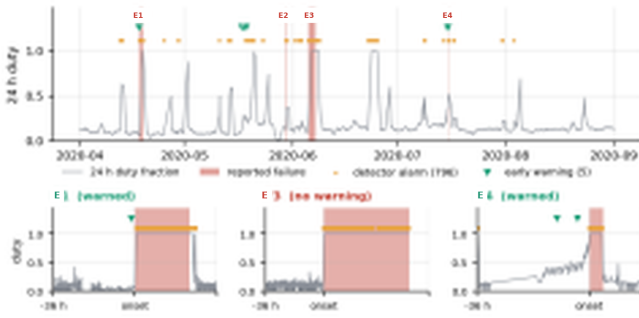


Figure 4: The delivered system across the five-month test period. Amber marks detection alarms, green triangles early warnings, red bands reported failures.

6 EFFICIENCY AT SCALE

Efficiency here is measured, because a monitoring system that cannot keep pace with its own stream is undeployable and 1.5 M points is enough for the question to bite.

Table 7 settles the deployment question with room to spare, though the figure worth quoting is the end-to-end one. Online scoring alone costs 0.008 ms per window, the 0.12 s detection-scoring row plus the 0.006 s CUSUM row divided across the 16,773 windows, which is a margin of roughly 10^8 ; that number covers only the cheapest stage. Charging the full pipeline, pre-processing included, against the windows it produces gives 16.25 s for 16,773 windows, or 0.97 ms per window, a margin of about 10^6 against the 900,000 ms window period. That total is the six pre-processing stages of Table 7 at 15.61 s, plus window feature extraction at 0.52 s and detection scoring at 0.12 s; the k -means and Isolation Forest fits are one-off training costs that are not repeated per window and are excluded. We report the smaller figure as the headline because it is the one that describes the deployed system, and because 10^6 already settles the question. Note that the checkpoint led with the scoring-only figure and this report leads with the end-to-end one. Both are correct; only the emphasis has changed, since the end-to-end figure describes the deployed system.

The binding constraint is resampling and gap flagging at 7.18 s, with CSV parsing close behind at 5.19 s; both are ingestion costs, and a columnar store removes most of them in production. Drift-adaptive weekly refitting costs 2.2 s in total across the test period, which eliminates any efficiency argument against retraining often. Nothing here needs a central service; the system would run on constrained hardware at the vehicle.

7 OVERALL PROJECT ANALYSIS

7.1 What worked

Clustering was the highest value per hour spent. A 0.07 s k -means fit recovered the fault regime with no supervision and validated the entire feature set. It also supplied the gate, though cluster membership on its own matches the gated model’s F_1 and exceeds its PR-AUC; see Section 7.3.

Physically scoped features beat capacity. Reducing 41 features to 14 raised F_1 from 0.312 to 0.546. Every stage of this project rewarded domain reasoning over model sophisti-

Table 7: Measured performance over the full record, single machine. Rows are additive. The end-to-end figure of 16.25 s covers pre-processing, window feature extraction and scoring; the two model-fitting rows are one-off training costs and are excluded.

Stage	Time
CSV load, 1,516,948 rows	5.19 s
Resample and gap flagging	7.18 s
Offset correction	0.12 s
Median filtering, 7 channels	2.09 s
Digital debouncing, 8 channels	0.60 s
Frozen-channel detection	0.43 s
Window feature extraction	0.52 s
k -means clustering, $k = 2$	0.07 s
Isolation Forest training	0.26 s
Detection scoring, all windows	0.12 s
CUSUM early warning, all windows	0.006 s
Pre-processing throughput	97,190 rows/s
Online scoring cost per window	0.008 ms
End-to-end cost per window	0.97 ms
Online-only headroom	$\sim 10^8 \times$
End-to-end headroom	$\sim 10^6 \times$

cation.

Flagging gaps instead of filling them. 909 hours of missing record never became fabricated training data. This cost 17.65% of grid rows and was worth it.

7.2 What did not, and what it taught

The cycle as modelling unit. It failed because failure eliminates cycling. We demoted it to a feature source, which was at the time a salvage operation and not a plan; that the resulting cycle-count feature turned out to carry the early signal was luck as much as design. The useful version of this lesson is narrower than it looks. We were wrong about the grain, and we happened to be wrong in a direction that left the underlying quantity intact. Had the fault suppressed cycling without a preceding rise, the same correction would have produced nothing.

Isolation Forest over the full feature set. It lost to a one-line threshold. We could have retuned until it won and reported only that; diagnosing drift and dilution as the causes taught us more than a marginal win would have.

Engineered precursor features. Ten carefully motivated features made lead-time prediction slightly worse. The lesson concerns sample size rather than feature design: with four events, added dimensionality buys variance.

7.3 Planned analyses not delivered

The project’s stated ethos is to report what was found including failures, and that obliges us to be as explicit about analyses quietly dropped as about methods that visibly failed. Seven commitments from the proposal and checkpoint were not delivered.

SHAP attribution [12] was a headline proposal deliverable, framed as “an explanation, not just a flag”. It was cut when the delivered detector became an Isolation Forest gated by cluster membership over 14 features, for which the gate

and the duty feature dominate any attribution so completely that per-window SHAP values would have restated the model definition back to us instead of explaining individual alarms. That is a defensible reason to cut it and a poor reason to have promised it; attribution would have been genuinely useful had the supervised lead-time model shipped, which is where it should be reinstated.

The three named ablations (cycle-level versus raw-sample features, analogue-only versus analogue-plus-digital, and trailing-window length) were scheduled for week 7 and never run. Feature scoping from 41 to 14 is a related result, though it is none of the three. This is the clearest instance of scope lost to the two course corrections.

Peak memory and serialised model size were listed as efficiency metrics and are absent from Table 7, which reports only time.

Association rules with support and confidence over digital flag co-occurrence were proposed and never attempted, having been displaced when the modelling unit changed.

The seasonal-naïve persistence baseline was named as a must-beat and was missing from the checkpoint; it is included in Tables 3 and 4 here, where it performs poorly (F_1 0.297 at 267 false alarms per 30 days).

Comparison at a fixed alert budget was promised, omitted from the checkpoint, and is delivered in Section 5.2 above.

The cluster-0-alone comparison was promised at the checkpoint, where cluster membership used on its own matched the gated model at F_1 0.58 and beat it on PR-AUC (0.40 against 0.37). The checkpoint stated the gated model would be justified in week 7 on the grounds that a continuous score can be traded against an alert budget while a binary assignment cannot. Section 5.2 supplies exactly that argument for the learned score generally, but the head-to-head against cluster-0 alone was not run, and it is the comparison most likely to have changed the shape of the delivered system. Table 3 carries no cluster-0 row for that reason: the two figures we have, F_1 0.58 and PR-AUC 0.40, come from the checkpoint’s model comparison and not from a test-period confusion matrix, and we would rather leave the row out than report precision and recall we did not measure.

None of these changes a conclusion, but listing them is the same discipline applied to the three methods that failed outright.

7.4 How roadblocks redirected the project

Data veracity issues did not merely complicate the work, they redirected it. The 909 hours of missing record forced the gap-flag design; the H1 offset trap forced a physically guarded cleaning rule and nearly destroyed a channel; the frozen-sensor ambiguity would have discarded 3% of the record on a naïve reading. Most consequentially, label incompleteness, measured at two thirds of the continuous-run regime, capped precision and shifted the project’s centre of gravity away from detection, which the labels could not reward further, toward early warning, where the question was still open.

Table 8: Planned eight-week schedule (21 June to 15 August 2027) against what happened. Both course corrections came out of results. Neither was a response to running short on time.

Wk	Dates	Planned vs actual	Status
1	21–27 Jun	EDA, cycle segmentation	as planned
2	28 Jun–4 Jul	Cleaning, feature store	as planned
3	5–11 Jul	Clustering	as planned, $k = 2$
4	12–18 Jul	Anomaly detection. Cycle unit abandoned; 15-min windows adopted. IF lost to a physical threshold.	correction 1
5	19–25 Jul	Supervised lead-time model attempted. Ten precursor features built; no improvement.	attempted
6	26 Jul–1 Aug	Model and decide. Switch rule fired; CUSUM fallback shipped. LSTM dropped on evidence.	correction 2
7	2–8 Aug	Efficiency and ablations. Efficiency delivered; three named ablations cut (Section 7.3).	partial
8	9–15 Aug	Write-up and repository	as planned

7.5 Plan versus execution

Table 8 sets the original schedule against what actually happened. Weeks 1 to 3, covering 21 June to 11 July, ran to plan. The two deviations both occurred at decision points where a result contradicted an assumption, and in both cases the schedule absorbed the change because week 8 was deliberately left unallocated at proposal time. The one genuine shortfall is week 7, where the ablation programme was cut.

7.6 Loose ends

Not everything in this project resolved into a lesson, and two questions are still open.

We never established why the rolling 30-day reference underperformed. It reached $F_1 = 0.463$ against 0.546 for the same model on a fixed reference, which is the opposite of what a drift diagnosis predicts. The explanation offered in Section 5.1, that some of the drift is within-reference, is a plausible guess and nothing more; we did not test it, and a reviewer should treat it as unresolved. It is possible the weekly refit interval is simply wrong, or that 30 days is too short a window at this cycling rate.

The engineered precursor features are a second loose end. They were built directly from Finding 1, they measure the quantity that Finding 1 says carries the early signal, and they made the model slightly worse. We attribute this to sample size, which is the most likely explanation at four events, but we cannot demonstrate it, and it remains possible that the self-referencing normalisation discarded exactly the absolute-level information the model was using.

7.7 Limitations

Four events is the binding statistical constraint on every supervised claim. Results derive from a single compressor on one metro system and may not transfer. Precision is bounded by label completeness. The early-warning layer warns on half the events. All four constrain the result and all four are stated here.

8 FUTURE WORK

Ordered by expected effect on the answer:

Instrument cycle timing at source. The strongest early signal is cycle frequency, which we recovered indirectly from a 10-second digital flag. A dedicated cycle-event log would sharpen it substantially at near-zero cost, and is the single change most likely to move early-warning performance.

Pool more events. Every supervised limit here is a four-event limit, not a method limit. MetroPT-1 and MetroPT-2 [18] cover different vehicles and include oil-leak faults; pooling them would both raise the positive count and test cross-fault generalisation.

Deploy a real-time dashboard. At 0.008 ms per window the scorer is effectively free. A streaming deployment surfacing regime assignment, anomaly score and cycle-rate CUSUM to the maintenance desk would close the loop to operations, and the latency budget is already established.

Test transfer across fleets. Whether cycle-frequency features generalise across compressor models and transit systems, or whether each unit requires its own baseline, is an open and practically decisive question for anyone operating a fleet.

Move toward remaining useful life. Detection tells a planner something is wrong; a calibrated RUL estimate tells them how long they have, which is what schedules are actually built from. That requires the additional events above before it is feasible.

9 CONCLUSION

Between 21 June and 15 August 2027 this project built a complete data mining pipeline across 1,516,948 sensor readings, reducing them to 16,773 analysable windows in 16.25 s end to end and delivering a two-layer monitoring system.

The finding that matters most is not a metric. It is that this compressor fails in three stages, that cycling frequency rises 20% before onset and collapses 95% during a fault, and that duty cycle, the obvious and most-used indicator, does not move until the fault has already arrived. That single fact determines what any future early-warning system for this class of machine must measure.

The delivered detector achieves $F_1 = 0.580$ at recall 0.985, catching all four documented events with five missed windows in total, and the full pipeline runs about six orders of magnitude inside real-time budget end to end. The early-warning layer warns on two of four events at 6.1 hours mean lead and 0.4 false alarms per 30 days. Detection is solved; prediction is half solved.

Two planned methods failed and were replaced. A pre-registered switch rule fired, and we honoured it, though Sec-

tion 5.3 shows that call was closer than we would have liked. The project's most transferable lesson is methodological: at every stage, physical reasoning about this machine outperformed additional model capacity, and the constraints that ultimately bound the result were the quantity of labelled events and the granularity of the sensors. Algorithmic sophistication was never what stood in the way.

REFERENCES

- [1] M. Barros, B. Veloso, P. M. Pereira, R. P. Ribeiro, and J. Gama. Failure detection of an air production unit in operational context. In *IoT Streams for Data-Driven Predictive Maintenance and IoT, Edge, and Mobile for Embedded Machine Learning*, pages 61–74. Springer, 2020.
- [2] C. Bergmeir and J. M. Benítez. On the use of cross-validation for time series predictor evaluation. *Information Sciences*, 191:192–213, 2012.
- [3] V. Chandola, A. Banerjee, and V. Kumar. Anomaly detection: A survey. *ACM Computing Surveys*, 41(3):1–58, 2009.
- [4] M. C. Comstock, J. E. Braun, and E. A. Groll. A survey of common faults for chillers. *ASHRAE Transactions*, 108:819–825, 2002.
- [5] N. Davari, B. Veloso, R. P. Ribeiro, P. M. Pereira, and J. Gama. Predictive maintenance based on anomaly detection using deep learning for air production unit in the railway industry. In *2021 IEEE 8th International Conference on Data Science and Advanced Analytics (DSAA)*, pages 1–10. IEEE, 2021.
- [6] N. Davari, B. Veloso, G. d. A. Costa, P. M. Pereira, R. P. Ribeiro, and J. Gama. A survey on data-driven predictive maintenance for the railway industry. *Sensors*, 21(17):5739, 2021.
- [7] J. Gama, I. Žilobaite, A. Bifet, M. Pechenizkiy, and A. Bouchachia. A survey on concept drift adaptation. *ACM Computing Surveys*, 46(4):1–37, 2014.
- [8] J. Gama, R. P. Ribeiro, and B. Veloso. Data-driven predictive maintenance. *IEEE Intelligent Systems*, 37(4):1–2, 2022.
- [9] C. R. Harris et al. Array programming with NumPy. *Nature*, 585:357–362, 2020.
- [10] J. D. Hunter. Matplotlib: A 2D graphics environment. *Computing in Science & Engineering*, 9(3):90–95, 2007.
- [11] F. T. Liu, K. M. Ting, and Z.-H. Zhou. Isolation forest. In *2008 Eighth IEEE International Conference on Data Mining*, pages 413–422. IEEE, 2008.
- [12] S. M. Lundberg and S.-I. Lee. A unified approach to interpreting model predictions. In *Advances in Neural Information Processing Systems 30*, pages 4765–4774, 2017.
- [13] W. McKinney. Data structures for statistical computing in Python. In *Proceedings of the 9th Python in Science Conference*, pages 56–61, 2010.
- [14] E. S. Page. Continuous inspection schemes. *Biometrika*, 41(1/2):100–115, 1954.
- [15] F. Pedregosa et al. Scikit-learn: Machine learning in Python. *Journal of Machine Learning Research*, 12:2825–2830, 2011.
- [16] P. J. Rousseeuw. Silhouettes: A graphical aid to the interpretation and validation of cluster analysis. *Journal of Computational and Applied Mathematics*, 20:53–65, 1987.
- [17] T. Saito and M. Rehmsmeier. The precision-recall plot is more informative than the ROC plot when evaluating binary classifiers on imbalanced datasets. *PLoS ONE*, 10(3):e0118432, 2015.
- [18] B. Veloso, J. Gama, R. P. Ribeiro, and P. M. Pereira. MetroPT: A benchmark dataset for predictive maintenance. Zenodo, 2022. <https://doi.org/10.5281/zenodo.6854240>.
- [19] B. Veloso, R. P. Ribeiro, J. Gama, and P. M. Pereira. The MetroPT dataset for predictive maintenance. *Scientific Data*, 9(1):764, 2022.